Energy Infrastructure
Challenge Provider

U.S. DEPARTMENT of ENERGY | Office of Technology Commercialization

GLOBAL INDUSTRY CHALLENGE 2026

Global Industry Challenge 2026

Phase 2 Challenge Description

CHALLENGE PROVIDER:
U.S. Department of Energy Office of Technology Commercialization

FOCUS AREA:
Energy Infrastructure

TITLE:
Quantum-Enhanced Strategic Siting of Energy Storage and Microgrids for the Era of AI and Industrial Load Expansion

1. INTRODUCTION

Welcome Message

Welcome to the 2026 Global Industry Challenge on Quantum-Enhanced Strategic Siting of Energy Storage and Microgrids. This challenge invites teams to apply near-term quantum computing methods to one of the most consequential planning problems in the U.S. power system today: where to deploy storage and microgrids to absorb the rapid growth in AI data center loads, advanced manufacturing demand, and electrification, while preserving reliability and resilience under increasingly volatile weather and contingency conditions. Your work will help shape how grid planners, utilities, and federal partners evaluate emerging quantum tools for high-stakes infrastructure decisions.

Overview of the Organization

The U.S. Department of Energy Office of Technology Commercialization (DOE OTC) advances the transition of federally developed and federally relevant technologies into deployed solutions that strengthen U.S. economic and energy security. OTC works across the National Laboratory complex, federal R&D programs, and industry partners to accelerate commercialization pathways for emerging capabilities, including quantum information science, advanced energy systems, and resilient infrastructure. This challenge is run in partnership with Connected DMV (CDMV) and the Global Industry Challenge consortium, and it builds on DOE-aligned priorities for grid modernization, large-load integration, and the practical evaluation of quantum methods for high-impact public-sector problems.

2. THE CHALLENGE

Problem Statement

Develop a quantum or hybrid quantum-classical method that improves how grid planners decide where and at what capacity to deploy battery energy storage systems (BESS) and microgrids across a transmission or distribution test system. Solutions should explicitly handle (a) multi- scenario uncertainty in load and generation, (b) contingency and resilience constraints (N-1 or weather-driven outages), and (c) the combinatorial complexity introduced by integrating large new flexible loads, including AI data centers and industrial expansion. Teams will benchmark their quantum approach against an established classical solver on a published grid test system and document where the quantum or hybrid method meaningfully changes solution quality, scenario coverage, runtime behavior, or robustness.

Background and Context

U.S. electricity demand is entering a structural growth period for the first time in roughly two decades. AI data center campuses, semiconductor fabs, electrified transport, and reshored industrial loads are arriving faster than transmission infrastructure can be permitted and built. Storage and microgrids are increasingly central to bridging this gap because they can be sited and energized on shorter timelines than new high-voltage lines. The placement decision is hard: it is a large mixed-integer program with nonlinear AC power flow physics, contingency constraints, and a scenario tree that grows quickly when planners attempt to capture realistic weather and outage variability. Classical formulations rely on linearization, scenario reduction, and decomposition heuristics, and run-time often constrains how thoroughly planners can explore the design space.

Quantum optimization research over the past several years has produced credible candidate methods for combinatorial planning problems of this shape, including QAOA on QUBO formulations, variational quantum eigensolvers for power flow, and analog neutral-atom encodings of independent-set and graph problems. Several peer-reviewed studies have applied these methods to unit commitment, optimal power flow, and transmission expansion at small scales. The frontier question now is whether hybrid quantum-classical workflows can produce decision-relevant improvements on realistic siting and sizing problems within current NISQ device limits, and whether the results scale meaningfully toward planner-grade systems.

Desired Outcomes

Winning teams will deliver a working Phase 2 prototype that maps a well-defined siting and sizing decision onto a quantum or hybrid optimization workflow, runs that workflow on a simulator or available NISQ device, and benchmarks the result against a classical baseline on the same test system. Successful submissions will: clearly state the planning question being answered, document the encoding from grid problem to quantum problem, justify the algorithm and platform choice, demonstrate at least one credible quantitative comparison against a classical solver, and provide a defensible scaling estimate for Phase 3 hardware and qBraid resource needs. Submissions are expected to be honest about where quantum methods help, where they do not, and what would have to be true at the hardware level for the approach to become planner-relevant.

3. RESOURCES AVAILABLE

Key Datasets

Teams are encouraged to build on publicly available, citable data so that judges can reproduce results. Recommended sources include:

- IEEE PES open test systems, including the IEEE 9,14, 30, 34, 39, 57, 68 and 118-bus systems and the RTS-GMLC and RTS-96 test cases for resilience-aware planning (note, IEEE 9, IEEE 39, and IEEE 68 bus cases are widely used for distribution studies).
    - Please contact Grid-Q at Oak Ridge National Lab (ORNL) for additional datasets
- NREL ReEDS scenario outputs and the NREL Wind Integration National Dataset (WIND Toolkit) and National Solar Radiation Database (NSRDB) for renewable generation profiles.
- EIA Form 860 (generator-level inventory) and Form 923 (generator operations and fuel use) for U.S. fleet composition.
- DOE Form OE-417 historical electric disturbance events and the FEMA National Risk Index for hazard exposure layered onto candidate siting locations.
- Open data portals from PJM, MISO, ERCOT, CAISO, and SPP for hourly load, locational marginal prices, and interconnection queue data, useful for representing AI data center and large-load buildouts.
- Sample data center load profiles from published EPRI and LBNL studies and NREL data center research, suitable for proxy modeling of campus-scale AI load.

Where teams wish to use additional proprietary or curated datasets, the Aqora platform supports dataset hosting (https://aqora.io/datasets). Down-selected teams may receive access to a curated planning-grade case study compiled by the Challenge organizers, distributed under a non-commercial research license.

Software and Tools

- Classical optimization and power systems: Pyomo, JuMP and PowerModels.jl, PandaPower, MATPOWER, GridLAB-D, and PSS/E-style open equivalents. Solvers: Gurobi or CPLEX academic licenses, HiGHS, SCIP, or Ipopt for nonlinear baselines.
- Quantum frameworks: Qiskit and Qiskit Optimization, PennyLane, Cirq, OpenQAOA, Amazon Braket SDK, and the qBraid SDK and Lab. Teams targeting analog neutral- atom hardware should review Bloqade for QuEra Aquila workflows.

- Hybrid orchestration on qBraid: classical CPU and GPU runtimes are available alongside QPU access, which allows decomposition workflows where a classical master problem calls quantum subproblem solvers.
- Visualization and reporting: matplotlib, NetworkX, GeoPandas, and Folium for grid topology and siting heatmaps.

Additional Resources

- Ajagekar and You, "Quantum computing for energy systems optimization" (Energy, 2019) and follow-on hybrid quantum-classical applications in process and power systems planning.
- DOE Energy Storage Grand Challenge Roadmap and DOE Long Duration Storage Shot reports.
- EPRI work on storage integration and resilience metrics, and NERC reports on AI and large-load interconnection.
- Recent surveys on QAOA and quantum approximate optimization for power systems, including transmission expansion and unit commitment formulations.
- ORNL-led GRID-Q project's solution on unit commitment for IEEE 9-bus, 39-bus, and IEEE 118-bus cases.
    - https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11249834
- qBraid documentation at https://docs.qbraid.com/home/introduction for setup, credit usage, and platform-specific tutorials.

4. GUIDELINES AND EXPECTATIONS

Submission Requirements

Submissions must follow the standardized Phase 2 format: a maximum of **3 pages (PDF)**, excluding the required cover page template and references, using 11-point Times New Roman, single spacing, and submitted via Aqora. Only one submission is required per team.

The **official Global Industry Challenge cover page template is required** and must be included as the first page of the submission. Teams may not recreate, modify, or substitute this template. Submissions that do not comply with these requirements may be disqualified.

Your submission should address:

- Focus area and rationale
- Technical approach to quantum integration
- Stakeholder relevance
- Data modeling strategy
- Quantum platform justification and resource needs

Evaluation Criteria

Submissions will be scored using a rubric by reviewers including domain experts in power systems planning and in quantum optimization. Strong submissions will demonstrate technical correctness on both sides of the bridge: a credible grid problem formulation and a defensible quantum approach. Vague or ungrounded claims about quantum advantage, formulations that cannot be reproduced, and submissions that do not establish a classical baseline will score poorly.

Category	Description
1. Focus Area and Rationale	Justification of selected use case (storage siting, microgrid topology, capacity expansion, or co-optimization with large loads) and alignment with a measurable planning outcome.
2. Technical Quantum Integration	Soundness of the quantum or hybrid formulation: encoding, algorithm choice, integration with classical planning logic, and treatment of nonlinear power flow constraints.
3. Stakeholder and Industry Relevance	Identification of beneficiaries (utilities, ISOs, DOE, communities, large loads) and clarity on how the method improves a real planning decision.
4. Data Modeling Strategy	Use of public, reproducible grid test systems and weather or outage scenario data; transparency about candidate sets, scenarios, and assumptions.
5. Platform Justification and Resource Needs	Appropriateness of selected quantum hardware or simulator, with stated qubit, depth, shot, and classical compute estimates and a credible Phase 3 scaling plan.
6. Clarity of Communication	Structure, coherence, and completeness of the 3-page technical submission for a mixed audience of planners and quantum specialists.

Rules and Regulations

- Solutions must be reproducible. Code, data references, and run instructions must be sufficient for a third-party reviewer to verify the headline results in Phase 3.
- Teams must disclose any use of proprietary data and cannot submit results that depend on data the judges cannot access.
- Use of generative AI tools is permitted for code support and writing, but the technical contributions, formulations, and results must be the team's own work.
- Any siting or planning recommendations are research outputs and do not constitute investment, regulatory, or operational advice.
- Teams should be careful with grid topology and asset-level data that may be sensitive, and should default to publicly released test systems wherever possible.

Phase 2 Detailed Challenge Description

In **Phases 2 and 3** of the Global Industry Challenge, participants will move from conceptual design to applied execution, demonstrating technical sophistication and real-world relevance. Phase 2 focuses on designing and justifying early-stage quantum solutions tailored to each challenge domain. Winning teams are expected to deliver, where possible, functional prototypes or circuit designs, define data preprocessing or encoding workflows, and justify their choice of algorithms and execution platforms, whether simulators or NISQ hardware. Equally important is demonstrating the value of the quantum approach compared to classical alternatives. In addition, teams should provide their estimates of the technical requirements needed to scale their solutions in Phase 3, including the type and extent of quantum hardware and simulator usage required via qBraid's platform.

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Energy and Critical Infrastructure: Develop a hybrid quantum-classical method for strategic siting and sizing of energy storage and microgrids in a power system facing rapid AI and industrial load growth.

Phase 2: Prototype Development and Technical Foundation
In Phase 2, your team will develop a working proof-of-concept that applies quantum or hybrid quantum-classical optimization to strategic siting and sizing of energy storage and microgrids. The phase emphasizes both technical development and the articulation of your solution's planning relevance.

1. FOCUS AREA AND USE CASE SELECTION

Select one or more of the four primary focus areas, based on your team's expertise and the planning question you can address most credibly within Phase 2's timeline:

- **Storage siting and sizing for resilience and AI load integration:** decide where to place BESS and at what energy and power capacity, given a candidate buildout of large new loads and a defined contingency set.
- **Microgrid boundary and topology design:** identify which subsets of buses and feeders should be islanded as microgrids to maximize served critical load under defined outage scenarios.
- **Multi-scenario capacity expansion under contingency:** co-optimize storage and microgrid investments across a representative scenario tree of weather, load, and outage realizations.
- **Co-optimization with large flexible loads:** integrate AI data center or industrial load flexibility (curtailment, time-shifting, on-site generation) into the siting decision so that storage and microgrid placements account for load-side controllability.

Justify your selection by tying it to a measurable planning outcome: improved resilience metric (e.g., expected unserved energy under N-1 or weather scenarios), capital efficiency, integration speed for new loads, or solution robustness across scenarios.

2. QUANTUM-DRIVEN SOLUTION DESIGN

Develop an early-stage proof-of-concept using quantum or hybrid quantum-classical methods tailored to your selected use case. Possible approaches include:

- **QAOA or VQE on QUBO and Ising formulations** of binary siting decisions and sizing buckets, with continuous variables (e.g., flow, dispatch) handled by an outer classical loop. Typical separation between quantum and classical formulation happens using Bender's decomposition or variations of it.
- **Quantum-enhanced decomposition:** Benders, Dantzig-Wolfe, or progressive hedging structures where the master investment problem runs classically and quantum methods are used inside the subproblem layer for combinatorial substructures.
- **Analog neutral-atom encodings:** map siting graphs onto Maximum Independent Set or related problems on Rydberg arrays (QuEra Aquila) for combinatorial selection of candidate locations.
- **Quantum-assisted scenario generation:** variational or sampling-based quantum models to enrich the scenario tree for weather, outage, or load realizations, particularly in correlated tail-risk regions.

Describe how the quantum component connects to the rest of the planning workflow: data preprocessing, encoding, classical pre- and post-processing, and the interface back to the planner-facing decision (which bus, what capacity, under what scenario coverage). Provide a one-figure system diagram of the hybrid architecture.

3. DATA MODELING AND SIMULATION

Build your prototype on a public, reproducible test system. Recommended starting points are the IEEE 14, 30, or 118-bus systems for tractable Phase 2 work, and the RTS-GMLC system for resilience-focused case studies. Augment the base topology with: (a) a candidate set of storage and microgrid locations, (b) a representative load growth overlay (for example, one or more synthetic AI data center loads of the order of 50 to 500 MW added to selected buses), and (c) a contingency or weather scenario set drawn from public sources such as NSRDB, the WIND Toolkit, or DOE OE-417 historical disturbance data.

Document your data modeling approach explicitly: the size of the candidate location set, the number of scenarios, the contingency definition, the encoding into your quantum problem (number of binary variables, problem graph structure, Hamiltonian construction), and the assumptions used to simplify any nonconvex AC power flow physics. Be transparent about what is preserved from the original planning problem and what is approximated to make the quantum encoding tractable.

4. QUANTUM PLATFORM AND RESOURCE PLANNING

Justify your platform choice from the Phase 3 supported list. Available platforms include:

- Amazon Braket SV1, DM1, TN1
- Fire Opal error mitigation software with IBM hardware
- IBM Eagle r3, Heron r1, Heron r2
- IonQ Aria-1, Forte-1
- IQM Garnet
- NVIDIA GPUs with Qiskit GPU, Cirq GPU, PennyLane Lightning, and cuQuantum available out of the box
- qBraid QIR simulator
- QuEra Aquila
- Rigetti Ankaa-2, Aspen-M3

Indicative guidance for this challenge domain: gate-based QAOA prototypes on small grid problems will typically fit within tens of qubits and circuit depths in the low hundreds, suitable for state-vector simulation, NVIDIA GPU-accelerated simulation, or small NISQ runs on IBM Heron or IonQ. Larger combinatorial siting graphs may map naturally onto QuEra Aquila for analog neutral-atom MIS-style problems. Hybrid decomposition workflows benefit from the qBraid Lab's combined CPU, GPU, and QPU access. State your expected qubit count, circuit depth, number of shots, and the classical compute budget needed to support hybrid loops, and identify the integration points with your classical planning code.

5. STAKEHOLDER IMPACT

Identify who benefits from your solution and how. Likely stakeholder groups include: investor- owned and public-power utility planners making 5-to-20-year capital decisions; ISOs and RTOs running interconnection studies for AI and industrial loads; state energy offices and DOE programs evaluating resilience and grid hardening investments; community and tribal entities pursuing microgrids for critical-load resilience; and large-load customers (data centers, fabs, manufacturing) negotiating siting and behind-the-meter storage. Describe the specific decision your method improves, the magnitude of the improvement on the test system, and what would need to be true at the hardware level for a planner to use the tool in a real planning cycle.